

3.1 Quadratic Graphs and Equations

- x-intercept (roots) $\Leftrightarrow$ when $y=0$
  - $\Rightarrow$ if there are 2 roots, the vertex will be below the y axis, and the x value will be the midpoint of the roots.
  - $\Rightarrow$ Repeated root (single) vertex will only touch the x-axis at the vertex
  - $\Rightarrow$ No roots $\Rightarrow$ will not touch x-axis.
- y-intercept $\Rightarrow$ when $x=0$
  - shown by c
- Shape
  - $\Rightarrow$ Always a parabola
  - $\Rightarrow$ Based on whether x^2 is +ve/-ve. (U/∩)

Basic form: $ax^2+bx+c=y$, $a\neq 0$

Factorising Quadratic Expressions

A method considered when solving a quadratic. Expresses quadratic as the product of 2 linear factors:

$(x-a)(x-b)$

3.2: Completed Square form

An alternate expression of a quadratic

$ax^2+bx+c \equiv a(x+p)^2+q$

All quadratics have a line of symmetry, through its vertex ($\frac{dy}{dx}=0$)

Vertex: $(-p, q)$, line of symmetry: $x=-p$

Nature of vertex is determined by the coefficient of x^2. (if it is $\cup$ / $\cap$)

Solving Equations:

$a(x-p)^2+q$ can then be solved by setting y to 0.

3.3 Quadratic Formula.

Can be derived by completing the square on a general quadratic expression. It can determine any roots:

$$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

The discriminant can be used to find the amount of roots on a quadratic

$b^2-4ac > 0$, 2 real roots

$b^2-4ac = 0$, One repeated root

$b^2-4ac < 0$ No real roots.